

001 — Technology Baseline

Six technology lines bound the orbital economy's investment cases, each with a hard physical or economic floor. Launch cannot fall below the propellant floor — yet launch is 12.3% of revenue at the company that dominates it. Orbital compute was out-chosen, not out-built.

Constitution pin	1.2.0
As of	2026-09-18
Universe	SPCX 9% · RKLБ 5% · FLY 3% · LUNR 2% · PL 2% · KRMN 3% · VOYG 2% · YSS 3% · HAWK 1% · IRDM 5% · GSAT 3% · SATS 4% · LHX 2% · NOC 2% · LMT 2% · BA 2% · RTX 2% · HWM 2% · MRCY 2% · BWXT 2% · TDG 1% · HEI 1% · WWD 1% · CW 1% · KTOS 1% · AVAV 1% · TER 1% · GOOG 6% · MSFT 6% · NVDA 4% · VRT 4% · UTHR 7% · MRK 3% · BMY 3% · AMGN 2%
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Executive Summary

44 ARTIFACTS 35 tickers · 6 lines	74 GRADED CLAIMS 65 demonstrated · 5 modeled · 4 claimed	3 / 6 LINES HOLD 3 hold · 0 falsified · 3 unresolvable	12.3% LAUNCH SHARE of SPCX revenue — and falling
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Headline output: launch is **12.3% of revenue at the company that dominates launch**, and its self-reported throughput **fell** — mass to orbit **–25.6%**. Orbital compute was not out-built; it was **out-chosen** by the one actor with cheap orbital access, which deployed **1.4 GW** of terrestrial compute instead.

Across 35 tickers and 44 artifacts, the disclosure record is consistent with — and in two places decisively contradicts — the thesis's founding premise that launch is the orbital economy's centre of gravity. **The physics bounds all survive. The value story does not.**

- SUPPORTED

Line 1 — launch cost floor. The only disclosed cost-per-kilogram figure is \$14,667/kg, **14.7× above** the threshold. But launch volume at the launch monopoly fell, and A1b of the constitution is falsified on the issuer's own metrics.
- SUPPORTED

Line 2 — orbital power envelope. No listed issuer discloses orbital-compute revenue on any of four readings. SPCX's AI-infrastructure revenue is adjacent — and terrestrial on every axis.
- INDETERMINATE

Line 3 — constellation economics. The binding constraint is **fixed-cost absorption**, demonstrated at seven issuers. The falsifier's prose census is **unbuilt**, not unavailable.
- SUPPORTED

Line 4 — microgravity economics. No listed pharma discloses commercial microgravity manufacturing. The economic test — beating a **72.0%** incumbent gross margin — cannot be run at all.
- INDETERMINATE

Line 5 — orbital compute. The numerator does not exist anywhere; the denominator is fully public. The ratio is not large, it is **undefined**.
- INDETERMINATE

Line 6 — spectrum and slots. The premise is demonstrated four ways, including **\$856M paid in cash**. The falsifier needs a registry outside the corpus.

Method note. Every quantitative claim carries an evidence grade. Three of the six falsifiers cannot fire from any public source, and **the three failures have three different causes** — an unbuilt census, a number that exists nowhere, and an unreachable registry. Distinguishing them is more useful than one "insufficient data" label, because each has a different remedy.

The Register, and the first two lines

Line	Governing bound	Value	Constraint	Verdict
1 — Launch cost floor	Propellant mass × price ÷ payload	46.0 USD/kg	MASS_LAUNCH_COST	SUPPORTED
2 — Orbital power envelope	Array area per MW BOL	5,080 m ² /MW	POWER	SUPPORTED
3 — Constellation economics	Break-even revenue multiple	1.69×	MANUFACTURING_RATE	INDETERMINATE
4 — Microgravity economics	Incumbent gross margin	72.0%	DEMAND	SUPPORTED
5 — Orbital compute	Rejection area per MW at 300 K	2,419 m ² /MW	THERMAL	INDETERMINATE
6 — Spectrum and slots	Reference transaction, three blocks	\$19.6B	REGULATORY_SPECTRUM	INDETERMINATE

Line 1 — Launch cost has a hard floor

```
metric=demonstrated_price_per_kg_to_LEO_P50 threshold=1000 op=< basis=ANY_OF_A_B_C
```

Reusability removes the hardware, not the propellant: 4,600 t at \$1–2/kg is \$4.6–9.2M per flight, or **\$46–92/kg** at 100 t to LEO. Rocket Lab discloses both a price and a cost — **\$30,333/kg on basis A and \$14,667/kg on basis B**, each DEMONSTRATED, **14.7–30×** above threshold. **But the verdict turns on a grade distinction, not a settled distance:** the disclosed bases span \$2,939 to \$30,333/kg, while SPCX's marginal cost models at **~\$525–875/kg — below the threshold**, MODELED not demonstrated, and the falsifier says *demonstrated* price. **Verdict: supported on every disclosed basis; NOT EVALUABLE on the one where it would fire.**

Line 2 — Orbital power and thermal rejection set a hard envelope

```
metric=listed_issuer_orbital_compute_revenue_disclosed threshold=0 op=>
```

AMO irradiance at 30–32% cell efficiency with eclipse (×1.587), packing (÷0.85) and degradation (÷0.90) gives **5,080 m² of array per MW — 2.1×** the naive 2,449 m² cell-area figure, reconciling against an independently published ~5,640 m²/MW. **Zero of five issuers disclose orbital-compute revenue** on any reading. **Verdict: supported — on a definitional test, not on absence:** SPCX's adjacent AI revenue fails DA-20's four-way reading, since nameplate compute draw counts GPUs in data centers. **Reading it as orbital compute will wrongly fire this falsifier.**

FALSIFIER EVALUATION · LINES 3 AND 4

Lines 3–4 — fixed-cost absorption and the microgravity hurdle

Line 3 — Manufacturing rate binds, not launch capacity

```
metric=share_of_named_issuers_citing_launch_availability_as_primary_delay_cause threshold=0.5
op=>
```

The bound. Not a cost but a ratio: revenue must absorb a fixed operating base at the observed gross margin. Planet Labs earned a **53.5% gross margin** — the best of any space *operator* here — and still lost 37.1% at the operating line, because opex was 90.6% of revenue. Break-even is **1.69× current revenue**. At YSS's 24.0% gross margin the same arithmetic requires **2.86×**.

Falsifier result: unresolvable, and the reason matters. The risk-factor language **is** filed and public. What does not exist is any **structured field for delay cause**. The census is **unbuilt**, not unavailable — bounded work across 19 filings, and the only one of the three with a remedy inside the platform.

Verdict: indeterminate — and the adjacent finding is stronger than the falsifier. The measured constraint is fixed-cost absorption, visible in XBRL without reading a word of risk-factor prose.

Recommendation: re-scope this falsifier to the break-even multiple, which is computable and comparable across issuers.

The YSS detail, at line level. Revenue \$92.547M, cost of revenue \$70.367M, gross profit \$22.180M at **24.0%**; operating expenses \$63.493M, or **68.6% of revenue**; operating result **\$(41.313)M**; R&D \$5.766M, or 6.2%. **The fixed cost base is 2.9× the gross profit** — which is the whole story. A 24% gross margin is a real, positive manufacturing spread; the company is below minimum efficient scale, and **manufacturing cost is not the binding constraint — manufacturing volume is**. Those are different claims with different remedies. *Caveat: Q2 2026 includes the All.Space acquisition, so the expense ratio may be inflated by deal costs; segment confirmation was not available from the extract.*

Line 4 — Demonstrated physics, unproven economics

```
metric=listed_pharma_microgravity_commercial_manufacturing_disclosure threshold=0 op=>
```

The bound. The hurdle is the incumbent process's return. Amgen: \$10,054M revenue less \$2,811M COGS gives **72.0%** gross. Bristol Myers: **71.3%**. **Two independent buyers agree to within 0.7 points** — that convergence is the number an orbital process must clear.

Falsifier result. Zero of four listed pharma disclose commercial microgravity manufacturing.

Verdict: supported on the disclosure falsifier, unresolvable on the economic test. Value-per-kg and cost-per-kg for any microgravity product **do not exist publicly** — a stronger condition than an unreachable registry, because here the number exists nowhere. **The line holds on the weaker of its two tests, and the register says so.**

FALSIFIER EVALUATION · LINES 5 AND 6

Lines 5–6 — orbital compute and allocated spectrum

Line 5 — Does orbital compute close before terrestrial wins?

```
metric=orbital_to_terrestrial_cost_per_kW_ratio threshold=3 op=< basis=ANY_OF_A_B_C
```

Falsifier result: unresolvable, on one side only. The **denominator is fully public** — Microsoft's capex and capacity guidance give a terrestrial cost per kW on all three bases. **The numerator does not exist.** A threshold of 3 is unreachable when the numerator is undisclosed: **the ratio is not large, it is undefined.** Reporting one basis alone would fail the contract; reporting three with a missing numerator is the honest state.

Verdict: indeterminate. The numerator and denominator fail for **different** reasons, and the register keeps them distinct.

Line 6 — Finite allocated resources, priced as assets

```
metric=new_entrant_primary_spectrum_or_slot_grant_without_incumbent_acquisition threshold=0
op=>
```

The bound. The only recent transaction pricing allocated spectrum is SPCX–EchoStar at **\$19.6B** for AWS-4, H-Block and AWS-3 — carried as a **transaction value**, not a unit price, because the per-MHz-POP denominator is not disclosed and a bare ratio would fail the contract.

The premise is demonstrated four ways. EchoStar's quarterly operating margin runs **2.3% → 5.7% → 460.5% → 118.1% → 10.7%** — two quarters carrying the spectrum gain, the rest the actual operating business at 2–11%. That is roughly **\$27B of gains across 2025 H2** against **\$15.0B of full-year revenue**: the asset was worth $\sim 1.8\times$ the business holding it. SPCX's 10-Q shows **\$856M of spectrum payments in cash** — a headline price is an assertion, **a cash-flow line is a payment**. Iridium is profitable at 15.1% on licensed L-band. **Grading: the \$856M is filed — (FACT).** The **\$27B is an inference from GAAP operating income whose classification the filing does not break out — the direction is certain, the figure is (DEDUCTED).**

Two caveats, and they limit the premise's reach. First, **this is a sell-side observation, not a buy-side one** — EchoStar realised the value by **exiting** spectrum, so it demonstrates the exit, not that **holding** is durable. Second, the buy-side counterexample cuts: Iridium's operating margin **fell 8.1 points** while revenue grew 3.8%, because **a durable licence does not confer pricing power in the service layer above it.**

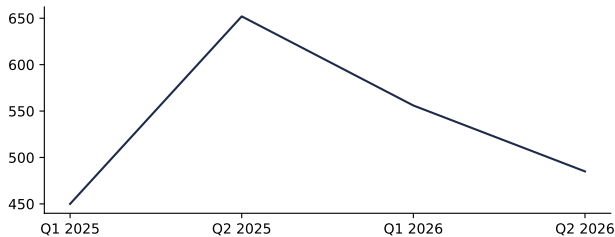
Verdict: indeterminate. The falsifier needs FCC IBFS and the ITU Space Network List — public, outside the reachable corpus. **The premise does not need the falsifier, and the falsifier does not weaken the premise.**

HEADLINE FINDING · THE LAUNCH MONOPOLY IS SHRINKING

Launch volume is falling where launch is concentrated

SPCX's Q2 2026 10-Q reports its own chosen throughput metrics. They fell. This is the most consequential datum in the thesis, because it runs opposite to the premise the universe was built on.

Key business metric	Q2 2026	Q2 2025	Change	Source
Mass to orbit (t)	485	652	-25.6%	10-Q 0001628280-26-052535
— customer payloads	87	88	-1.1%	same
— internal payloads	397	563	-29.5%	same
Falcon launches	37	45	-17.8%	same
— customer launches	10	9	+11.1%	same
— internal launches	27	36	-25.0%	same
Starship launches	1	1	H1: 3 → 1	same



Mass to orbit by quarter, metric tons. Q2 figures are as filed; **Q1 figures are derived** as the filed half-year total less the filed Q2 — $1,102 - 652 = 450$ and $1,041 - 485 = 556$.

SPCX calls mass to orbit "**a key indicator of SpaceX's capacity and scalability.**" It fell 25.6%, and the contraction is concentrated in **internal** launches — the Starlink deployment demand that was supposed to justify the cadence. Customer launches also fell across the half, from 21 to 17.

Constitutional consequence: A1b — "launch is the master **value** variable" — is **falsified on the issuer's own metrics**. A1a — launch cost is the master **cost** variable — survives; the floor still bounds every downstream case.

One distinction favours the launch business. SPCX's Space segment carries a **65.8% gross margin** — the highest of any segment here, though group-level margins run higher (TDG 59.4%, UTHR 87.3%) — with cost of revenue **flat** while revenue rose 29.0%. The \$(542)M quarterly segment loss is **Starship development R&D** at \$1,076M, 111.9% of segment revenue. **Falcon launch economics are good; this is a funding decision about one rocket.**

Launch is 12.3% of revenue at the launch company

The company the universe was built around is not primarily a launch company, and its own segment disclosure says so.

Segment	Q2 2026 revenue	Share	Operating margin	H1 2026 change
Connectivity (Starlink)	\$4,291M	54.9%	+38.6%	+49.1%
AI (Grok, X, compute)	\$2,561M	32.8%	not disclosed	largest growth contributor
Space (launch, Dragon)	\$962M	12.3%	-56.3%	-1.9%
Consolidated	\$7,814M	100%	-1.8%	+53.7%

The H1 comparison is the cleaner one. H1 2025 predates the xAI merger, so the AI column is absent — but the **Space column is directly comparable across both halves, and it fell 1.9% while total revenue rose 53.7%.** Any thesis using SPCX as the launch-sector read-through is reading **12% of the company and 0% of its growth.**

The reporting entity changed mid-series

An X merger (2025-03-28), an xAI merger (2026-02-02, common control), the largest IPO in history in June 2026 — 638.9M shares at \$135.00, **net proceeds \$85,675M** — a five-for-one split, and a **\$60B all-stock Cursor acquisition** pending. Common-control accounting recasts prior periods, so **any growth rate quoted across those boundaries mixes real growth with entity change.** The Space-segment series is the one that survives the boundary, and it is the one that fell.

Starlink, read correctly: subscribers **doubled** to 12.0M while ARPU fell **22.4%** to \$66/month. Growth is being bought with price — and the **enterprise, government, aviation and maritime half grew more than the consumer half** (\$939M vs \$764M). The consumer story is the weaker half.

Orbital compute was out-chosen, not out-built

1.4 GW NAMEPLATE COMPUTE Q2 2026 · up from 0.4 GW	+250% YEAR OVER YEAR 3.5× its own year-ago figure	\$21.5B H1 CAPEX INCREASE data centers named first	\$60B CURSOR, PENDING all-stock · closing Q3 2026
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The H1 2026 capital-expenditure increase of **\$21,511M** is attributed, verbatim, to **"the build out of data centers and related infrastructure, and space launch facilities and related infrastructure" — data centers named first** — and the company states it *"expect[s] to allocate a significant amount [of launch capacity] to our AI segment."*

Two more signals from the same filing

- **The only new risk factor this quarter is about terrestrial data centers.** Item 1A carries exactly one addition, headed *"Risks related to our AI infrastructure and data center operations."* Not launch. Not Starship. Not spectrum. **The market's mental model of SPCX and the company's own risk disclosure have diverged.**
- **F1's power bound is confirmed as a live constraint — on the ground.** That risk factor names **"power constraints"** among data-center threats, and the forward-looking list includes *"our ability to obtain sufficient power, GPUs."* The power constraint is real, binding, and **the first company to hit it is hitting it terrestrially.**
- **And the thermal bound has already been observed in flight.** Starcloud-1 carried an NVIDIA H100 and **cannot run it at full power because cooling capacity is insufficient** — F2 observed rather than derived, and the first empirical data point in the thesis. *Graded CLAIMED: press report, private company.*

The bound. Radiative rejection follows $P/A = \epsilon \cdot \sigma \cdot T^4$: 413 W/m² at 300 K, or **2,419 m² of radiator per MW**, falling to 313 m² at 500 K — a **7.7× compression**. **The array is temperature-independent and the radiator is not**, so total deployed area runs 7,499 m²/MW on solar against **313 m²/MW on nuclear at 500 K**. Removing the array entirely produces that **24× reduction** — the concrete quantitative case for a space reactor. **The array, not the radiator, is the dominant area term**, which is why Line 2 reports the full DA-15 temperature range rather than assuming one value. **And the terrestrial gap decides this line: 7,499 m² of deployed surface per MW against roughly 200–400 m² of building footprint — 20–35× more**, which no cost curve bends. At the F5 propellant floor the ~50 t/MW is **\$2.3–4.6M per MW** in launch cost alone.

The conclusion is not that orbital compute was out-competed. It is that it was out-chosen by the only party able to choose it — with \$21.5B of disclosed capital expenditure, and a preference that is terrestrial.

Margin rises with distance from programme risk

Across the eight issuers where the ladder can be tested end to end, the ordering is monotone in one variable and runs **opposite** to the universe's implicit ordering — best at the component suppliers (TDG 44.8%, TER 32.9%), worst at a single-buyer operator (GSAT 7.4%).

Issuer	Role	Operating margin	Revenue growth
TDG	aftermarket-weighted components	44.8%	+22.5%
TER	semiconductor test equipment	32.9%	+103.9%
HWM	aerospace components, R&D 0.3%	27.9%	+24.1%
HEI	flight-critical parts, aftermarket	25.5%	+25.3%
VRT	data-centre thermal management	19.5%	+24.1%
KRMN	composites, payload fairings	19.1%	+58.2%
LMT	prime	12.4%	+10.5%
KTOS	growth defense tech	0.35%	+30.5%
MRCY	rad-hard electronics	0.03%	—
FLY	launch, spacecraft	−80.9%	+657%

Three suppliers agree to within 0.4 points despite unrelated businesses — Vertiv 19.5%, Curtiss-Wright 19.3%, Karman 19.1%. A band that tight is what a **structural determinant** looks like. Four primes sit inside a **2.3-point** band (10.1–12.4%) with very different space exposure, so **prime margin carries no information about space leverage**.

The ordering is inverted relative to narrative — the names the market treats as growth exposure earn the least: Kratos **0.35% on +30.5% growth**, Mercury **0.03%**, every space operator negative. **The inverse is not absolute**: HEICO, TDG and Vertiv all pair real margins with real growth — but they are incumbents selling into installed bases, not the names the growth narrative points at.

The binding constraint is the expense base. Planet Labs holds the best operator gross margin at **53.5%** against a 90.6% expense ratio and still loses 37.1% at the operating line — **not unit economics, but scale relative to a fixed base**. R&D intensity separates the regimes: FLY 60.8%, PL 35.5%, RKL 35.2% against HEI 2.6%, TER 2.9%, **BWXT 0.5%**.

The buyers are 73× the sellers — and unpersuaded

The microgravity thesis is usually framed as a supply problem. The financials invert that framing: the buyers are not capital-constrained, they are the largest R&D spenders on earth.

73× DEMAND VS SUPPLY \$39.6B/qtr vs \$2.2B/yr cohort	72.0% HURDLE RATE incumbent gross margin	7.3× MERCK R&D ALONE vs the whole cohort's revenue	0 / 4 PHARMA DISCLOSING microgravity manufacturing
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Merck, Bristol Myers and Amgen turn over **\$39,634M in a single quarter**; the seven pure-play space companies combined turn over roughly **\$2,170M per year**. Merck spends **\$15,789M on R&D annually** — **7.3× the entire cohort's revenue**, and one Merck quarter alone is **4.5× that cohort's annual revenue**.

Why the capital does not move. Those R&D ratios are not steady-state research — they are **acquired IPR&D**, the accounting home of buying pipeline assets; Amgen's 12.2% equity-to-assets ratio, matching Globalstar's 12.0%, is the balance-sheet consequence. **Growth strategy and the microgravity thesis are substitutes for the same capital, and the growth strategy wins.** Equity cushions span 32.3% / 25.5% / 12.2% (Merck / Bristol Myers / Amgen), so **adoption should appear first at the less-levered buyers**.

Monopsony caps the supplier, and the installed base escapes it. Buyer concentration alone is not the discriminator — the primes' buyers are the same DoD that buys HEICO's parts. **The discriminator is whether revenue spreads across programmes or depends on one winning. Globalstar is the purest monopsony here and earns the thinnest operator margin, 7.4% — one buyer, one price, revenue falling 3.5%, and non-operating items at 4.6× its operating result.**

The counter-weight: Karman's capital structure **absorbs 60% of its operating income** — \$34.829M operating against \$14.032M net. **The first issuer here where the below-the-line bridge is more material than the operating line**, so operating-margin rankings overstate levered suppliers.

DA-23 is a whole-statement sign defect

The extracted financial block strips the sign from negative values. This was recorded as an operating-income defect. **It is not — it affects every signed line.**

gross profit \$84.576M - operating expenses \$142.090M = -\$57.514M (arithmetic) reported

OperatingIncomeLoss = +\$57.514M

Line at RKLБ, Q2 2026	Extract shows	What it must be	Basis
Operating income	+\$57.514M	-\$57.514M	component identity
Net income	+\$49.258M	-\$49.258M	below-the-line bridge unavailable
EPS (diluted)	+\$0.08	-\$0.08	follows net income

Why net income must be flipped. If the operating line is truly a \$(57.514)M loss, a positive net income would require **+\$106.772M of below-the-line income in one quarter**. Rocket Lab holds \$2.129B of cash — roughly \$21M per quarter at a generous 4% — and carries \$152.4M of debt. **\$106.8M is not available to it.**

Why this is worse than previously recorded: net income is the input to P/E, ROE and every common screening factor. **A loss-making issuer reads as profitable on any screen built from the metrics block.** The blast radius is the whole valuation toolchain, not one line.

Six issuers confirmed flipped — SPCX, YSS, RKLБ, FLY, BA, PL — against **23 of 23 profitable issuers clean**. Four detectors exist, in descending reliability: the component identity; the gross-profit bound (Voyager fails it by \$46,951M); the below-the-line bridge; and margin plausibility. **The EPS × shares test is unreliable**, passing on both sides of a flip at RKLБ, FLY and Voyager.

Period and classification defects

DA-26 and DA-27 — two period defects, one value and one label

DA-26, annual figures mislabelled as quarterly, is universal at 23 of 23 issuers. It is a **whole-statement failure**: at Amgen the Q4 row carries the annual total for **both** revenue and operating income, confirmed at TER, CW and VRT — **a model keyed on the Q4 row overstates both lines by roughly 4×**. It lands on the year-end quarter at non-calendar issuers, the subtlest form. **Scope caveat**: the rows are reliably *not a quarter*, but not reliably *a full year* — WWD's FY2025 row implies a year-end quarter that **dips 16% sequentially mid-sequence**, which no other issuer's annual reading requires.

DA-27, labels derived from the calendar quarter, is 4 of 4. The platform buckets a period by calendar quarter from 1 January, then labels it with the issuer's fiscal year — exact for December year-ends, off by one otherwise. **Four non-calendar issuers, four offsets; every December issuer, none.** A pattern that separates a population perfectly on one variable is a **mechanism, not a coincidence**. The two defects compound at non-calendar issuers.

DA-24 — a second mechanism, with an identical consequence

EchoStar's 2025 Q3 and Q4 operating income **is not operating**: a spectrum-licence sale was classified such that it flows through the operating line, producing margins of **460% and 118%**. **Distinct from DA-23** — this defect is **classification**, not sign — but the consequence is the same, and so is the remedy: **verify operating income against its components before use**. The single rule covers both mechanisms across three issuers. **Any screen ranking EchoStar's operating income is comparing a licence sale against operating businesses.**

Three extracts cannot support a baseline

These are **platform coverage holes, not issuer defects**, and all three are excluded from rankings on input-availability grounds rather than on their economics.

Ticker	What is wrong	Consequence
AVAV	net_income_loss null in every row; revenue null in five of ten	all three detectors fail
HAWK	four non-agreeing share counts (4.2M to 98.0M); balance sheet 1.85× stale	EPS bridge fails by 72% post-IPO
VOYG	operating income \$51.408M against gross profit \$4.457M	fails the gross-profit bound

DA-28 — a listing invalidates share-count detectors. Around an IPO, weighted-average share counts straddle two capital structures, so **EPS × shares cannot reconcile on either side** — and unlike DA-23, no arithmetic test recovers the true figure. **A DA-23 screen by EPS reconciliation will produce false positives on recent listings.**

The safe set is narrow. Gross margins, balance-sheet ratios and component identities survive. **Everything computed from net income, EPS, or a quarterly series must be rebuilt from the filing.**

Five findings the thesis was not looking for

- **Margin runs inverted to narrative.** Component suppliers earn roughly **2×** the primes that buy from them; the capital-equipment supplier earns **3×**; the growth names earn nothing. **No issuer here offers both growth and margin except TER and Karman.**
- **Fixed-cost absorption binds, confirmed at seven issuers.** The best gross margin in the sector still loses 37% at the operating line.
- **Orbital compute was out-chosen, not out-built** — by the only actor with cheap orbital access, which deployed 1.4 GW terrestrially and named **power constraints** in its only new risk factor.
- **The demand end is 73× the supply end and unpersuaded.** The barrier is willingness against a 72% incumbent margin, not capacity.
- **Half the critical inputs are structurally unpriced** — and the clearest case is named: **space-grade solar cells are a two-supplier market, SolAero (RKLB) and Spectrolab (BA)**, supplying the component whose efficiency is the **first term in bound F1**, with one leg a subsidiary of a launch competitor. **But it is structural, not priced** — neither discloses those units separately. The same applies to liquid engines, solid motors and rad-hard electronics: Lockheed reports Space as a named segment and still aggregates the components away, and ULA is equity-method and invisible at both parents.

RESOLUTION PATH · WHAT WOULD CHANGE EACH VERDICT

The six lines and the disclosure that would move each

The open questions here are **disclosure questions, not physics questions** — **three** of the six bounds derive from constants (F1, F2, F5), the rest from filings and one transaction.

LINE 1 — LAUNCH COST FLOOR · HOLDS

A customer list price or a fully-loaded cost at commercial cadence. **No issuer discloses either. Cheaper launch does not fire this falsifier — price is not cost.**

LINE 2 — ORBITAL POWER ENVELOPE · HOLDS

Orbital-compute revenue on any of four readings. **SPCX's AI-infrastructure revenue is adjacent and fails the four-way reading** — a documented false-positive trap.

LINE 3 — CONSTELLATION ECONOMICS · UNRESOLVABLE, UNBUILT CENSUS

A coding of Item 1A and MD&A delay language across 19 issuers. **The language is filed; nothing structures it** — the only line with a remedy inside the platform.

LINE 4 — MICROGRAVITY · HOLDS ON THE FALSIFIER, UNRESOLVABLE ON THE ECONOMICS

A value-per-kg or cost-per-kg figure for any microgravity product. **Neither exists** — Varda is private; UTHR's partnership is too small to force a segment.

LINE 5 — ORBITAL COMPUTE · UNRESOLVABLE, NUMERATOR ABSENT

An operator-disclosed orbital cost per kW. **The denominator is public; the numerator is disclosed by nobody.**

LINE 6 — SPECTRUM AND SLOTS · UNRESOLVABLE, CORPUS GAP

A query of FCC IBFS and the ITU Space Network List for primary grants to entities with no prior US holdings. **Public, and outside the reachable corpus.**

Method and its limits

Evidence grading. Every claim is graded DEMONSTRATED, CLAIMED or MODELED — 65 / 5 / 4 across 44 artifacts. Much of the baseline is necessarily **modeled**: nobody reports radiator mass or cost per kilogram. **Citations** are **SEC accessions as plain text** — the corpus carries no viewer links, so inventing one would be fabrication.

EVIDENCE · CROSS-COHORT FACT TABLE

Evidence across all cohorts — eighteen issuers

Badge legend: FACT = DEMONSTRATED · DEDUCTED = MODELED · VIEW = CLAIMED.

Ticker	Governing metric	Value	Grade
SPCX	Mass to orbit, Q2 2026	485 t vs 652 t, -25.6%	FACT
RKLB	Per-launch cost — the only disclosure	\$14,667/kg (basis B)	FACT
PL	Best space-operator gross margin	53.5% gross, -37.1% operating	FACT
KRMN	Supplier margin, spread across programmes	19.1% operating, +58.2% growth	FACT
FLY	Growth driver, named by the issuer	+657%, "Spacecraft Solutions"	FACT
YSS	Opex ratio at a sub-scale base	68.6% opex on 24.0% gross	FACT
SATS	Spectrum gains vs operating business	~ \$27B vs \$15.0B FY revenue	FACT
IRDM	The licence is durable, the service is not	15.1% , down 8.1 pts YoY	FACT
GSAT	Thinnest operator margin, single buyer	7.4% , revenue -3.5%	FACT
TER	Capacity seller, fungible end markets	32.9% operating, +103.9%	FACT
HWM	Component franchise benchmark	27.9% operating at R&D 0.3%	FACT
HEI	Highest margin in the supplier tier	25.5% operating, R&D 2.6%	FACT
CW	Pricing power on low volume growth	19.3% on +5.4% growth	FACT
LMT	Only prime with a named Space segment	12.4% — still aggregates	FACT
KTOS	Growth defense tech, margin collapse	0.35% on +30.5% growth	FACT
MRCY	The F4 gate's only pure-play	0.03% operating	FACT
MSFT	Terrestrial capacity added per year	\$115,948M capex → 2.9–11.6 GW	FACT
AMGN	Highest-margin buyer, most levered	72.0% gross, 12.2% equity	FACT

Four primes inside a 2.3-point band (10.1–12.4%) with very different space exposure — so prime margin carries no information about space leverage. Lockheed reports Space as a named segment and still aggregates the components away.

Nine amendment candidates, and what comes next

Every finding below requires owner approval before it changes the constitution. None is executed.

#	Candidate	Trigger
1	Split F5 into F5a (fully reusable) / F5b (partially reusable)	the propellant floor bounds Starship-class only
2	DA-23 — re-scope from one line to ALL signed lines	RKLB: three lines flipped in one quarter
3	DA-24 — operating income contaminated by asset-sale gains	EchoStar's 4.6× revenue quarter
4	New disposition class: unresolvable-from-platform	Line 6 needs registries the corpus cannot reach
5	Split A1 into A1a / A1b — highest priority	SPCX mass to orbit –25.6%; A1b falsified
6	DA-25 — normalised per-unit metrics not reproducible	RKLB's per-launch figure vs audited tables
7	DA-26 — annual mislabelled quarterly, 23 of 23	whole-statement period failure
8	DA-27 — labels derive from the calendar quarter, 4 of 4	perfect partition on fiscal year-end
9	DA-28 — IPO discontinuity breaks share-count detectors	HAWK fails the EPS bridge by 72%

What thesis 002 should do

1. **Build the delay-cause census** — 19 filings, the only unresolvable line with a remedy inside the platform.
2. **Cover Varda** — the demand-side economics exist nowhere in the listed universe.
3. **Resolve the three unusable extracts** before ranking anything on AVAV, HAWK or VOYG.
4. **Fix the four platform defects** — DA-23, 26, 27 and 28 corrupt every XBRL read, and DA-23's extended scope makes loss-makers read as profitable on any screen.
5. **Deepen the four highest-information names:** TER (capacity pricing), HEI (installed-base margin), PL (the best operator gross margin), and SPCX's Connectivity segment — the only profitable space-adjacent operating business at scale.